

Hamza Ahmad Mir

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EDUCATION

National University of Computer and Emerging Sciences (FAST)

Bachelor of Science in Computer Science

Islamabad, Pakistan

Aug. 2022 – Apr. 2025

Relevant Coursework: Design and Analysis of Algorithms, Artificial Intelligence, Parallel & Distributed Computing, Computer Networks, Operating Systems, Database Systems, Compiler Construction, Computer Architecture

EXPERIENCE

Computer Vision Intern – KDD Labs

Islamabad, Pakistan

Jul. 2025

- Developing and testing computer vision algorithms for image preprocessing and feature extraction.
- Working with deep learning models for visual data analysis using OpenCV and TensorFlow.
- Assisting in dataset preparation, annotation, and model evaluation.

Game Designer Intern – M-Labs

Islamabad, Pakistan

Jul. 2024 – Aug. 2024

- Designed and prototyped gameplay mechanics using Unity and C#.
- Collaborated with developers and artists to refine game feel and UI/UX; participated in playtesting and balancing.
- Gained hands-on exposure to the full game development pipeline, including scoping, iteration, and delivery.

Graphic Designer (Volunteer) – Literary, Art & Computing Societies

FAST NUCES, Islamabad

Aug. 2022 – Apr. 2025

- Built and maintained ECHO, a JavaScript-based web portal for student society management and event coordination.
- Managed digital presence and produced web-ready visual assets for society events and announcements.

PROJECTS

Web & Full-Stack Development

ECHO | JavaScript, HTML/CSS

- Web portal for student society management – event posting, member coordination, and announcements.
- Designed the full front-end interface and implemented core backend logic.

RetinaVerse | Python, Deep Learning, Web

- Web-based platform for AI-assisted retinal disease image analysis.
- Integrated a ResNet deep learning model with a browser front-end for clinician-facing diagnosis support.

CTF Platform | Python, Flask, HTML/CSS

- Intentionally vulnerable web application built for cybersecurity education.
- Implemented common OWASP vulnerabilities (SQLi, XSS, IDOR) in a controlled learning environment.

Desktop & Mobile Applications

Gym Trainer App | C#, SQL

- Full-featured desktop application for gym management, client tracking, and workout planning.
- Designed relational database schema and built a Windows Forms UI for administrators and trainers.

Secure-Chat | Python

- Client-server encrypted chat application ensuring confidentiality, integrity, and availability.
- Implemented cryptographic protocols for end-to-end message security.

AI & Machine Learning Applications

TumorScope | Python, Jupyter, ResNet18

- Deep learning tumor classification system with Grad-CAM visual explanations for model interpretability.

EmoTrack / *Python, Jupyter, OpenCV*

- Real-time facial emotion recognition system using webcam input and a trained CNN classifier.

Multi-Robot A* Pathfinding / *C++*

- Multi-agent pathfinding system using A* algorithm in dynamic environments with collision avoidance.

Game Development

Re:Sling / *Unity, C#*

- Hypercasual sling-based mobile game prototype built during a game jam at M-Labs.

Tetris & Space Shooter / *C/C++, SFML*

- Classic Tetris and arcade-style space shooter with OOP principles and real-time input handling.

Pacman (Assembly + Parallel) / *Assembly/MASM32, C/C++*

- Complete Pacman in x86 Assembly with basic AI; parallel version using multithreading (threads & mutexes).

SKILLS

Languages: JavaScript, Python, C/C++, C#, Java, HTML/CSS, Assembly (MASM32)

Web: React, Node.js, HTML/CSS, REST APIs, Flask

Frameworks & Tools: Git, Unity, OpenCV, TensorFlow, Jupyter Notebook, SFML

Databases: SQL, MongoDB

AI / ML: ResNet, Grad-CAM, CNN, deep learning pipelines

Parallel Computing: OpenMPI, OpenMP, SIMD (AVX/NEON), OpenCL

Soft Skills: Communication, Project Management, Presentation, Adaptability, Flexibility, Critical Thinking

Languages (Spoken): English (Very Good), Urdu (Native), Punjabi (Native)